



9/1/2026

N + - Network Fundamentals & Building Networks.

Wide Area Networks.

TASK 1.

- ❖ Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

➤ Phone to Instagram Server Connection:

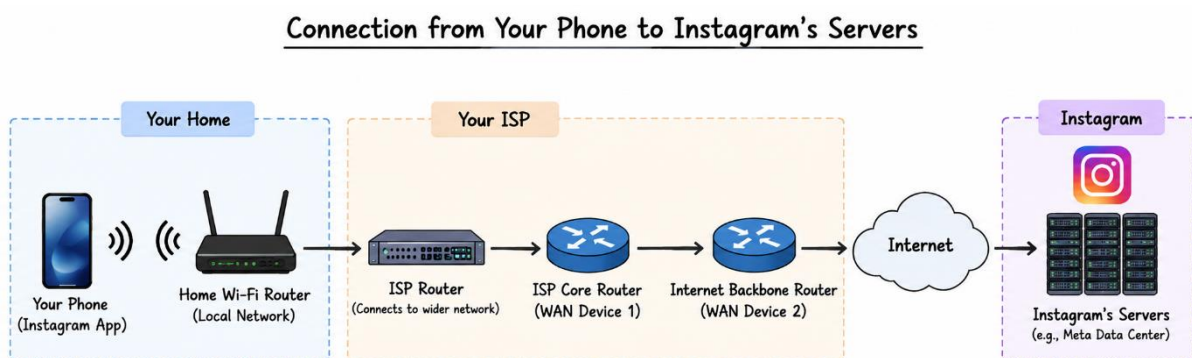


Figure : Connection from Your Phone to Instagram's Servers.

Explanation:

- ✓ The phone connects to the **Home Wi-Fi Router**, which forwards the request to the **ISP**. The data then passes through **two WAN routers** and the Internet before reaching **Instagram's servers**.

TASK 2.

- ❖ Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

➤ WAN Technologies Comparison:

WAN Technology	Speed	Cost	Real-Life Use
MPLS	High and reliable	High	Large offices like Zomato
Leased Line	Very high and dedicated	Very high	Organizations like IRCTC
Broadband Internet	Medium to high	Low	Homes and small offices

TASK 3.

- ❖ **List 4 Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop, Focus on how the VPN creates a private tunnel over the public WAN.**

- **VPN over WAN:**

A **VPN** creates a secure, encrypted tunnel over the public **WAN/Internet**. When using coffee-shop Wi-Fi, the device connects to the VPN server through this private tunnel, which protects the data from being easily seen by others on the public network. This allows apps like **Spotify or YouTube** to be accessed more securely.

TASK 4.

- ❖ **Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.**

- **WAN Technology for Flipkart:**

Suggested Technology: MPLS

Reasons:

- **Reliable connection:** MPLS provides a stable and reliable connection between warehouses in Ahmedabad, Mumbai, and Delhi.
- **Better traffic management:** It can efficiently handle large amounts of data between different warehouse locations.

Conclusion: MPLS would be a suitable WAN technology for connecting Flipkart's warehouses.